Derivation of Optimal Test Set for Detection of Multiple Missing-Gate Faults in Reversible Circuits

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Abstract— Logic synthesis of reversible circuits has received considerable attention in the light of advances recently made in quantum computation. Implementation of a reversible circuit is envisaged by deploying several special types of quantum gates, such as *k*-CNOT. Although the classical stuck-at fault model is widely used for testing conventional CMOS circuits, new fault models, namely single missing-gate fault (SMGF), repeated-gate fault (RGF), partial missing-gate fault (PMGF), and multiple missing-gate fault (MMGF), have been found to be more suitable for modeling defects in quantum *k*-CNOT gates. This article presents an efficient algorithm to derive an optimal test set (OTS) for detection of multiple missing-gate faults in a reversible circuit implemented with *k*-CNOT gates. It is shown that the OTS is sufficient to detect all single missing-gate faults (SMGFs) and all detectable repeated gate faults (RGFs). Experimental results on some benchmark circuits are also reported.

Keywords — Missing-gate faults, quantum computing, reversible logic, testable design, universal test set.

1. INTRODUCTION

Reversible logic can be employed to design information lossless circuits, and therefore, has the potential of reducing power consumption drastically [1-2]. It provides a basis for the newly emerging paradigm of quantum computing [3]. Reversible logic may offer a viable solution in the future with newer technologies, e.g., spintronics. Reversible circuits have numerous applications to optical computing, cryptography, nanotechnology, DNA technology, and low-power CMOS design [3-5].

Since quantum gates or circuits satisfy "no-cloning" behavior, and are information lossless, they do not permit fan-out, and ought to have an equal number of inputs and outputs. An *n*-input, *m*-output Boolean function *F* is said to be reversible if and only if $m = n$, and *F* is one-to-one. A reversible combinational circuit should consist of only reversible gates, which themselves implement reversible functions; such gates need to be specially designed, e.g., by using controlled-not (CNOT), Feynman [6], Fredkin [7] or Toffoli [8] gates.

Recently, several researchers have studied the problem of fault modeling and testing of reversible logic circuits [16-22]. Universal testability of reversible logic circuits designed with *k*-CNOT gates under the stuck-at fault model (both single and multiple) has also been investigated [20]. However, new fault models, namely, SMGF, RGF, PMGF, and MMGF, have been found to be more suitable for modeling defects in quantum *k*-CNOT gates [17, 22]. A *k*-CNOT based circuit can be implemented using trapped-ion technology, where the ions interact with laser pulses [3, 22]. Determination of universal test set for fault detection in reversible circuits has been studied for these fault models [17, 20]. The problem of detecting multiple missing-gate faults (MMGFs) by integer linear programming has been investigated [22]. It is shown earlier that in an $(n \times n)$ reversible circuit implemented with *k*-CNOT gates, addition of only one extra control line along with duplication of each *k*-CNOT gate yields an easily testable design, which admits a universal test set of size $(n + 1)$ that detects all SMGFs, RGFs, and PMGFs [14].

In this work, it is shown that it is always possible to construct an optimal test set (OTS) for detecting all multiple missing-gate faults (MMGFs) in an $(n \times n)$ reversible circuit implemented with *k*-CNOT gates. This optimal test set (OTS) detects all single missing-gate faults (SMGFs) and all detectable repeated gate faults (RGFs) in the circuit.

2. PRELIMINARIES

Reversible Logic: A reversible function has equal number of inputs and outputs, and simply induces a permutation on the set of input vectors to produce an output vector. Therefore, given an output vector, its corresponding input vector can be uniquely restored. A non-reversible Boolean function can always be implemented by a reversible circuit after appropriately transforming it to a reversible one by adding garbage lines and reversible gates [14]. By *garbage* is meant the number of extra outputs required to realize the given function.

Example 1*:* The function $\{x, y \rightarrow xy\}$ denoting AND operation is not reversible. By adding one extra input and two outputs, a modified but reversible function $\{x, y, z \rightarrow x, y, z \oplus xy\}$ can be constructed. The AND function can be realized at the output $z \oplus xy$, by setting the input z to constant zero; the circuit will have two "garbage" outputs. A Toffoli gate [8] can be used to realize this function.

Reversible gates: The basic CNOT type reversible gates used for synthesis are the following (Fig. 1):

(i) 1-input/1-output NOT ($x_1 \rightarrow \bar{x}_1$);

(ii) 2-input/2-output controlled NOT (CNOT) gate: $(x_1, x_2) \rightarrow (x_1, x_1 \oplus x_2)$;

and (iii) 3-input/3-output Toffoli gate $(x_1, x_2, x_3) \rightarrow (x_1, x_2, x_1 x_2 \oplus x_3)$.

A *k*-CNOT gate has *k* control inputs $x_1, x_2, \ldots, x_k$ and one target input *t*. It maps the input vector $(x_1, x_2, \ldots, x_k, t)$ to the output

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vector ($x_1, x_2, ..., x_k, t \oplus x_1 . x_2 \ldots x_k$). Thus, a k-CNOT gate has k+1 inputs and k+1 outputs; the first k outputs follow the respective inputs, and it inverts the target at the $(k+1)^{th}$ output if and only if all the k control inputs are 1.

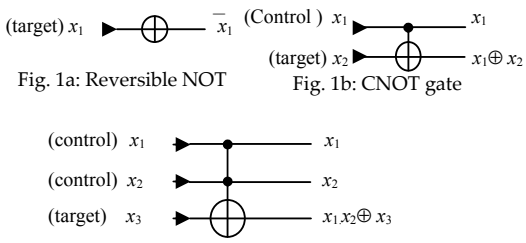

Fig. 1a: Reversible NOT

Fig. 1b: CNOT gate

Fig. 1c: Toffoli gate

***Fault model**:* Several new fault models for k-CNOT based reversible circuits were introduced earlier [17, 22]. These are single missing-gate fault (SMGF), the repeated gate fault (RGF), partial missing-gate fault (PMGF), and the multiple missing-gate fault (MMGF). In this section, we briefly explain with examples, the nature of these faults.

Single missing-gate fault (SMGF): This model corresponds to the case when one k-CNOT gate completely disappears from the circuit. In the presence of this fault, the CNOT gate behaves as a simple wire connection, i.e., the pulse implementing the gate operation is short, missing, misaligned or mistuned.

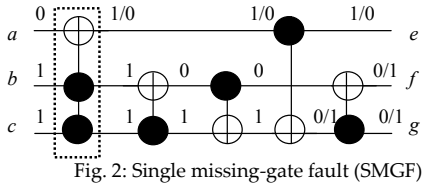

Fig. 2: Single missing-gate fault (SMGF)

Fig. 2 shows an SMGF marked by the dotted box in the reversible circuit ham3\design#1 benchmark, where the first 2-CNOT gate is missing. An SMGF is detected by setting logic 1 value on all the control inputs of the gate, and any value either 0 or 1 on the target input as well as on the wires not connected to the gate. In the example of Fig. 2, if we apply $\{a, b, c\} = \{0, 1, 1\}$ at the input of the circuit, the normal output would be $\{e, f, g\} = \{1, 0, 0\}$, whereas, in the presence of the SMGF fault marked by the dotted box, the output will be $\{e, f, g\} = \{0, 1, 1\}$. The number of possible *SMGFs* is equal to the number of gates in the circuit.

Repeated-gate fault (RGF): A repeated-gate fault *(RGF)* is an unwanted replacement of a k-CNOT gate by several instances of the same gate [22]. An RGF may be needed to model the occurrence of long or duplicated pulses. Fig. 3 shows an example, where first gate is repeated in the circuit ham3\design#1. The effect of this fault is thus same as that of an SMGF at the first 2-CNOT gate in the original circuit. If we apply $\{a, b, c\} = \{0, 1, 1\}$, the normal output would be $\{e, f, g\} = \{1, 0, 0\}$, whereas, in the presence of the above RGF marked by the dotted box, the output will be $\{e, f, g\} = \{0, 1, 1\}$. Hence, it is detected by the vector $\{a, b, c\} = \{0, 1, 1\}$.

It is clear that if a RGF replaces a gate by even number of instances of the same gate, its effect is similar to the effect of the SMGF with respect to the same gate. If the RGF replaces a gate by odd number of instances of the same gate, the fault is redundant, i.e., it does not change the function of the circuit. Further, it has been shown that any SMGF test set detects all detectable RGFs [22].

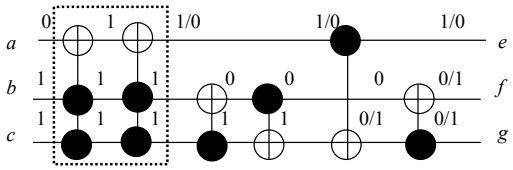

Fig. 3: RGF in ham3\design#1 reversible circuit

Partial missing-gate fault (PMGF): This is used to model the defects resulting from the partially misaligned or mistuned gate pulses [22]. It changes a k-CNOT gate into a p-CNOT gate, with $p < k$. The corresponding fault is called as *$(k - p)^{th}$ order* PMGF. Fig. 4 shows a first-order PMGF affecting the second control input of the leftmost gate. An SMGF can be seen as a 0-order PMGF. In the circuit of Fig. 4, if we apply $\{a, b, c\} = \{1, 0, 1\}$, the normal output would be $\{e, f, g\} = \{1, 1, 0\}$, whereas, in the presence of the first order PMGF fault as shown, the output will be $\{e, f, g\} = \{0, 0, 1\}$. Hence, the vector $\{a, b, c\} = \{1, 0, 1\}$ detects this fault. It has been shown [22] that a PMGF (of first or higher order) is detected when a 0 is applied to at least one of the affected control inputs and a 1 to all other control inputs. Thus, a higher order PMGF is detected by a test vector for a first order PMGF, the affected control input of which is one of those affected in the higher order PMGF. Hence, it is sufficient to consider first order PMGFs only.

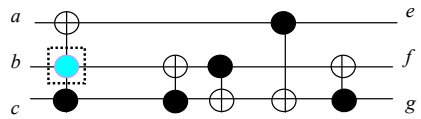

Fig. 4: PMGF in ham3\design#1

Multiple missing-gate fault (MMGF): This is defined as complete disappearance of two or more *consecutive* k-CNOT gates from the circuit.

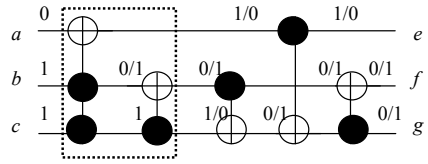

Fig. 5: MMGF in *ham3\design#1*

In the circuit of Fig. 5, it is shown that the circuit has an MMGF marked by the dotted box. This fault is detected by the vector $\{a, b, c\} = \{0, 1, 1\}$.

3. FAULT DETECTION IN A REVERSIBLE CIRCUIT

In this paper, we present an optimal test set (OTS) for detecting all MMGFs. No additional circuit overhead is required for testing. In the proposed algorithm, each gate is represented by a gate *Id*, which is used as a key to represent the permutation produced by the k-CNOT corresponding gate.

Given an n-input reversible circuit with k-CNOT gates all permutations of size n for gate *Id* are generated. In this algorithm, $\{N*(N+1)/2\}$ sets are constructed where the circuit depth is N. From the test patterns for these sets, a minimal cover is used to derive an optimal test set (OTS) for the reversible circuit.

A. Generation of Permutation

Each Toffoli gate with n inputs induces a permutation on the set $\{0, 1, 2, \ldots, 2^n-1\}$. In this paper, generation of Toffoli gates of size n means generation of permutations for all Toffoli gates of size n. The permutation of corresponding to each gate is stored in an array and the array index is considered as the gate *Id*. The input lines are numbered as $0,1,2,\ldots,(n-1)$ starting from the top. The selection of target line tl is made from line 0 to line $(n-1)$ one by one. After selecting the target line tl, the next task is to select the set of control lines from all possible sets of control lines with respect to the target line number tl. In other words, a total of
2^{n-1} sets of control lines with respect to the target line can be generated. We can use a binary number $b_{n-2}\ b_{n-3}\ \ldots b_1\ b_0$ to represent the corresponding control set. In other words, if a bit $b_i = 0$, then the corresponding line is not considered as a control line, whereas, if $b_j=1$, then the corresponding line is to be added as a control line of that control set.
We next apply all possible test patterns $\{0,1,2,\ldots\ldots,2^n-1\}$ and collect the corresponding set of output test pattern which we can be represented as a permutation:

$$G_i = \begin{pmatrix} 0 & 1 & 2 & \cdots & (2^n-1) \\ G_i(0) & G_i(1) & G_i(2) & & G_i(2^n-1) \end{pmatrix} = (G_i(0)G_i(1)\ldots G_i(2^n-1))$$

where $G_i(t)$ is the output test pattern of the input test pattern t and i is the identity number of that gate. In this way, all permutations for a Toffoli gates of size n are generated and stored in an array. Table 3 lists all Toffoli gates of size n=3 with gate *Id* and the corresponding permutation.

Table 3: Gate *Id*s and corresponding permutations of all Toffoli gate of size 3

Gate Ids	Corresponding Permutation
0	(4, 5, 6, 7, 0, 1, 2, 3)
1	(0, 5, 2, 7, 4, 1, 6, 3)
2	(0, 1, 6, 7, 4, 5, 2, 3)
3	(0, 1, 2, 7, 4, 5, 6, 3)
4	(2, 3, 0, 1, 6, 7, 4, 5)
5	(0, 3, 2, 1, 4, 7, 6, 5)
6	(0, 1, 2, 3, 6, 7, 4, 5)
7	(0, 1, 2, 3, 4, 7, 6, 5)
8	(1, 0, 3, 2, 5, 4, 7, 6)
9	(0, 1, 3, 2, 4, 5, 7, 6)
10	(0, 1, 2, 3, 5, 4, 7, 6)
11	(0, 1, 2, 3, 4, 5, 7, 6)

B. Optimal Test Set (OTS) for Detecting all Multiple Missing-Gate Faults (MMGFs)

Let us consider a reversible circuit RC of size n with depth N consisting of $g_0, g_{1,\ldots,} g_{N-1}$ reversible gates.
Let B_{ij} denote a block consisting of a sequence of gates from level i to level j of the circuit RC, i.e. $B_{ij}=\{g_i, g_{i+1,\ldots,} g_j\}$ for all $0 \le i \le j \le (N-1)$.

Let B_i^{pre} denote the predecessor block of B_{ij}, consisting of a sequence of gates from level 0 to level *(i-1)* of the circuit RC, i.e., $B_i^{pre} = \{g_0, g_1, \cdots, g_{i-1}\}\ \forall 0 < i \le N-1$.
Let B_j^{post} denote the next block of B_{ij}, consisting of a sequence of gates from level (j+1) to level (N-1) of the circuit RC, i.e., $B_j^{post} = \{g_{j+1}, g_{j+2}, \cdots, g_{N-1}\}\ \forall 0 \le j < N-1$.
Since there is no gate before the first gate of the circuit and after the last gate of the circuit, we have,
for $i = 0, B_0^{pre} = \{\} = \phi$ and in the case of
$j = N-1, B_{N-1}^{post} = \{\} = \phi$.
Let $P(B_{ij})$ denote the permutation of the block B_{ij} which is a product of $P(g_i), P(g_{i+1}), \ldots\ldots, P(g_j)$ where $P(g_k)$ denotes the permutation of the gate g_k for all $k = 0$ to $(N-1)$.
$B_0^{pre} = B_{N-1}^{post} = \phi$.
This implies that these two blocks contain no gate. If we apply any test pattern t as an input to these blocks, we will get the same output test pattern t. Hence, $P(B_0^{pre}) = P(B_{N-1}^{post}) = I$, where I is the identity permutation on the set $\{0,1,\cdots,2^n-1\}$.

Lemma 1: If an n-input reversible circuit RC with depth N is fault free (no multiple missing-gate fault is present) then, $g.f^{-1} = I$ where $g = P(B_i^{pre}).P(B_{ij}).P(B_j^{post})$, $\forall 0 \le i \le j \le N-1$ and $f = \begin{pmatrix} 0 & 1 & 2 & \cdots & (2^n-1) \\ f(0) & f(1) & f(2) & & f(2^n-1) \end{pmatrix}$ is a reversible specification of input-output test patterns of the circuit RC, and f^{-1} is the inverse permutation of f.

Proof: Let us assume a reversible circuit RC with n inputs and depth N consisting of $g_0, g_{1,\ldots,} g_{N-1}$ reversible gates.
So logically, $RC= g_0 \,.\, g_{1\cdot,\ldots} .g_{N-1}$ *(gate concatenation).* This implies that $P(RC)=P(g_0 \,.\, g_{1\cdot,\ldots} .g_{N-1})$ *and* $f=P(g_0).P(g_1)\ldots.P(g_{N-1})$

The total number of multiple missing-gate faults in an n-input reversible circuit with depth N is $\{N.(N+1)/2\}$. These $\{N(N+1)/2\}$ blocks of consecutive gates of the circuit RC may be missing from the circuit RC under multiple missing-gate fault, and are captured by B_{ij} where $0 \le i \le j \le N-1$. This implies that $N.(N+1)/2$ B_{ij} blocks correspond to all possible multiple missing-gate faults of the circuit RC. Since the circuit RC is fault free, i.e., no one gate is missing from any block B_{ij} for all $0 \le i \le j \le N-1$, we have:

- ⇨ $B_{ij}= g_i \,.g_{i+1\cdot\ldots} .g_j$
- ⇨ $P(B_{ij})=P(g_i \,.g_{i+1\cdot\ldots} .g_j)$
- ⇨ $P(B_{ij})=P(g_i)P(g_{i+1})\ldots P(g_j)$
- ⇨ $P(B_i^{pre}).P(B_{ij}).P(B_j^{post})$
  $= P(g_0).P(g_1).\cdots P(g_{i-1}).P(g_i).P(g_{i+1}).\cdots P(g_j).P(g_{j+1}).\cdots P(g_{N-1})$
  [Since, $B_i^{pre} = \{g_0, g_1, \cdots, g_{i-1}\}$ and
  $B_j^{post} = \{g_{j+1}, g_{j+2}, \cdots, g_{N-1}\}$]
- ⇨ $g=f$

⇨ $g.f^{-1} = I$

⇨ $g.f^{-1}(t) = t$, for any input test pattern *t* of the circuit *RC*.

Lemma 2 is useful in determining the test patterns for detecting all MMGFs in the circuit.

Lemma 2: If a multiple missing-gate fault occurs at the block B_{ij} of the circuit RC for some values of *i* and *j* ($0 \le i \le j \le N-1$), then $g.f^{-1} \ne I$, where B_{ij}^f is the faulty block corresponding to the block B_{ij} of the circuit *RC*, $g = P(B_i^{pre}).P(B_{ij}^f).P(B_j^{post})$ and *f* is the given reversible specification (permutation) of *RC*.

Proof: Since the block B_{ij} is missing, it means that the consecutive gates g_i, $g_{i+1,\ldots,}g_j$ corresponding to the block B_{ij} are missing from the circuit *RC*.

⇨ $B_{ij}^f = \phi$, where B_{ij}^f is faulty block corresponding to the block B_{ij}.

⇨ B_{ij}^f block does not contain any gate

⇨ P(B_{ij}^f)=I

If we assume that no block B_{ij} of the fault free circuit *RC* is redundant, then $P(B_{ij}) \ne I$.

Since *P(* B_{ij}^f *)=I* and $P(B_{ij}) \ne I$ for any i, j ($0 \le i \le j \le N-1$)

⇨ $P(B_{ij}^f) \ne P(B_{ij})$

⇨ $P(B_i^{pre})P(B_{ij}^f)P(B_j^{post}) \ne P(B_i^{pre})P(B_{ij})P(B_j^{post})$

⇨ $g \ne f$

⇨ $g.f^{-1} \ne I$ where *I* is the identity permutation

⇨ There exists at least one input test pattern *t* of the circuit RC such that $f^{-1}.g(t) \ne t$

Based on the above observations, we describe the following algorithm that generates an optimal test set (*OTS*) for detecting all possible MMGFs in a reversible circuit.

Algorithm:

Input: A reversible circuit *RC* of size *n* with depth *N* consisting of *k*-CNOT gates g_0, $g_{1,\ldots,}$ g_{N-1} .

Output: Optimal Test set (*OTS*) for detecting all multiple missing-gate faults of the circuit *RC*.

Step 1: Compute the reversible specification *f* of the circuit RC using $f = P(g_0).P(g_1)\cdots P(g_{N-1})$ and its inverse f^{-1}.

Step 2: for start_level = 0 to (*N*-1)

2.1. For end_level=start_level to (*N*-1)

2.1.1. Let us assume that a multiple missing-gate fault has occurred in the circuit RC between the start_level to end_level of the circuit RC; the B_{ij} block is missing from the circuit where *i*=start_level and *j*=end_level.

2.1.2. Compute the permutations $P(B_i^{pre})$ and $P(B_j^{post})$.

2.1.3. Compute the permutation g= $P(B_i^{pre})$. $P(B_j^{post})$, since $P(B_{ij}^f) = I$

2..1.4. Compute the final permutation corresponding to the block *Bij* as $P_{ij} = g.f^{-1}$

2..1.5. Decompose the final permutation P_{ij} into disjoint *k* (say) cycles as given below:

$P_{ij} = (n_1^1 n_2^1 \ldots n_{r1}^1)\cdot(n_1^2 n_2^2 \ldots n_{r2}^2)\ldots(n_1^k n_2^k \ldots n_{rk}^k)$, where n_j^i is the jth element of the ith cycle.

2.1.6. Construct a set $S_{Bij} = \{n_1^1, n_2^1, \ldots, n_{r1}^1, n_1^2, n_2^2, \ldots, n_{r2}^2, \ldots, n_1^k, n_2^k \ldots, n_{rk}^k\}$ from the *k* disjoint cycle elements corresponding to the block B_{ij}.

Step 3: Set all S_{Bij} as unmarked set ($0 \le i \le j \le N-1$)

Step 4: Initialize the optimal test set, $OTS = \phi$.

Step 5: We select a test pattern *t* among the unmarked sets S_{Bij} ($0 \le i \le j \le N-1$) which belongs to the maximum number of unmarked sets and mark those sets to which the selected test pattern *t* belongs.

Step 6: We continue Step 5 unless there is no such test pattern *t*, which belongs to more than one unmarked set S_{Bij} .

Step 7: If there is any unmarked set, then do the following:

7.1. For each unmarked set S_{Bij}

7.1.1. Mark the unmarked set

7.1.2. Take any element from that set and add to the set *OTS*.

Step 8: Finally, we get our desired optimal test set of test pattern, *OTS*.

Example 3 illustrates an MMGF which has occurred between the levels from 2 to 6 of the circuit *Circuit#1*.

Example 3: Consider a 4-input reversible circuit (*Circuit#1*) with depth 9 containing the sequence of gates $\{g_0, g_1, g_2, g_3, g_4, g_5, g_6, g_7, g_8\}$ whose corresponding gate *Id*s are {0, 8, 23, 8, 0, 1, 29, 30, 1}.

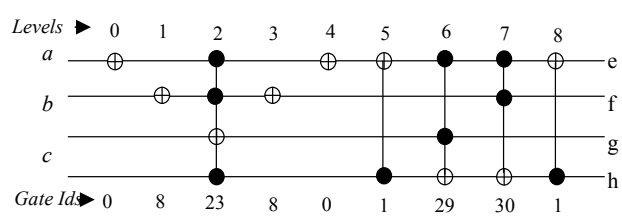

Fig.6: circuit1#1

Consider a multiple missing gate fault between the levels from *i* to *j* corresponding to the block B_{ij} where *i*=2 and *j*= 6; this means the $B_{2,6}$ block is missing from the circuit.

We first compute the reversible specification of the circuit *Circuit#1, f=P(g₀).P(g₁). P(g₂). P(g₃). P(g₄). P(g₅). P(g₆). P(g₇). P(g₈)* =(0,10,2,1, 4, 12, 6, 7, 8, 9, 3, 11, 5, 13, 14, 15)

Compute f^{-1}=(0, 3, 2, 10, 4, 12, 6, 7, 8, 9, 1, 11, 5, 13, 14, 15)

Since block $B_{2,6}$ ={23, 8, 0, 1, 29} is missing, then $B_{2,6}^f = \phi$.

This implies $P(B_{2,6}^{f}) = I$ where I is the identity permutation,

This implies g= $P(B_2^{pre}).P(B_{2,6}^{f}).P(B_6^{post})$, where $B_2^{pre} = \{0,8\}$ and $B_6^{post} = \{30,1\}$. P(B_2^{pre})=P(g_0).P(g_1)=(12, 13, 14, 15, 8, 9, 10, 11, 4, 5, 6, 7, 0, 1, 2, 3). P(B_6^{post})=P(g_7).P(g_8)=(0, 9, 2, 11, 4, 13, 6, 15, 8, 1, 10, 3, 5, 12, 7, 14)

Therefore, g=(5,12, 7, 14, 8, 1, 10, 3, 4, 13, 6, 15, 0, 9, 2, 11)

Compute $g.f^{-1}$=(12,5,7,14,8,3, 1, 10, 4, 13, 6, 15, 0, 9, 2, 11)

Decomposing it into disjoint cycles, we get 5 cycles:

$g.f^{-1}$ =(0, 12)(1, 5, 3, 14, 2, 7, 10, 6)(4,8)(9,13)(11,15)

Table 4: Different blocks and their corresponding sets for *ham3tc*

Set Indices	Levels From i	Levels To j	Block B_{ij}	g= $P(B_i^{pre})$. $P(B_j^{post})$	S_{Bij}
0	0	0	B_{00} ={3}	(0, 2, 1, 3, 7, 5, 6, 4)	{3,7}
1	0	1	B_{01} ={3,5}	(0, 3, 1, 2, 7, 4, 6, 5)	{1, 3, 5, 7}
2	0	2	B_{02} ={3,5,9}	(0, 3, 2, 1, 7, 4, 5, 6)	{1, 2, 3, 5, 6, 7}
3	0	3	B_{03} ={3,5,9,10}	(0, 3, 2, 1, 4, 7, 6, 5)	{1, 2, 3, 4, 5, 7}
4	0	4	B_{04} ={3,5,9,10,5}	(0, 1, 2, 3, 4, 5, 6, 7)	{1, 2, 3, 4, 7}
5	1	1	B_{11} ={5}	(0, 3, 1, 5, 7, 4, 6, 2)	{1, 3, 5, 7}
6	1	2	B_{12} ={5,9}	(0, 3, 2, 6, 7, 4, 5, 1)	{1, 2, 3, 5, 6, 7}
7	1	3	B_{13} ={5,9,10}	(0, 3, 2, 5, 4, 7, 6, 1)	{1, 2, 3, 4, 5, 7}
8	1	4	B_{14} ={5,9,10,5}	(0, 1, 2, 7, 4, 5, 6, 3)	{1, 2, 3, 4}
9	2	2	B_{22} ={9}	(0, 1, 2, 4, 7, 6, 5, 3)	{1, 2, 5, 6}
10	2	3	B_{23} ={9, 10}	(0, 1, 2, 7, 4, 5, 6, 3)	{1, 2, 3, 4}
11	2	4	B_{24} ={9, 10, 5}	(0, 3, 2, 5, 4, 7, 6, 1)	{1, 2, 3, 4, 5, 7}
12	3	3	B_{33} ={10}	(0, 2, 1, 7, 4, 6, 5, 3)	{3, 4, 5, 6}
13	3	4	B_{34} ={10, 5}	(0, 2, 3, 5, 4, 6, 7, 1)	{2, 3, 4, 5, 6, 7}
14	4	4	B_{44} ={5}	(0, 2, 3, 4, 5, 7, 6, 1)	{2, 4, 5, 7}

Therefore, $S_{B2,6}$= (0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15}

This implies that if we apply any test pattern from the above set, we can detect the multiple missing-gate faults corresponding to the block $B_{2,6}$. Example 4 illustrates how the proposed algorithm is used to derive an OTS for the above circuit.

In this example, the n = 3 and N = 5. The *ham3tc* reversible circuit can be represented as: RC= $g_0\, g_1\, g_2\, g_3\, g_4$, *where* $g_0 = 3$, $g_1 = 5$, $g_2 = 9$, $g_3 = 10$ *and* $g_4 = 5$. The reversible specification $f = P(g_0).P(g_1).\ P(g_2).\ P(g_3).\ P(g_4)$ = (0, 2, 1, 4, 7, 5, 6, 3).

The inverse permutation, f^{-1}=(0, 2, 1, 7, 3, 5, 6, 4). Table 4 shows the values of different blocks and their corresponding sets of the circuit *ham3tc*. Table 5 shows each step after deriving the sets corresponding to all MMGFs blocks needed in order to find the OTS set. Therefore, for the circuit *ham3tc*, *OTS* = {3, 5} or {3, 2}. The following lemmas (Lemma 3 and Lemma 4) justify that the optimal test set (OTS) is sufficient to detect all SMGFs and all detectable RGFs in the reversible circuit.

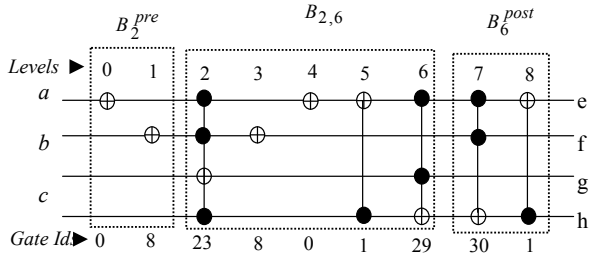

Fig.7: Diagram of circuit1#1 during MMGF at B_{26}

Example 4: Consider the *ham3tc* reversible circuit shown below:

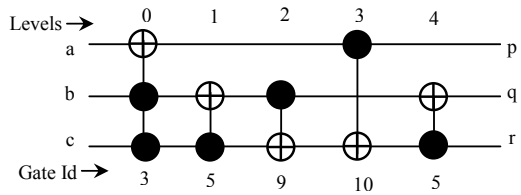

Fig. 8: the *ham3tc* reversible circuit

Table 5: All test patterns with their corresponding list of sets

Test pattern (t)	List of sets indices containing the test pattern *t*	Total no. of steps needed to derive *OTS*		
		Total no. of unmarked sets covered (Step1)	Total no. of unmarked sets covered (Step2)	Total no. of unmarked sets covered (Step3)
0		0	0	0
1	1,2,3,4,5,6,7, 8,9,10,11	11	1	0
2	2,3,4,6,7,8,9, 10,11,13,14	11	2	0
3	0,1,2,3,4,5,6, 7,8,9,10,11, 12,13	13	0	0
4	3,4,7,8,10,11, 12,13,14	9	1	0
5	1,2,3,5,6,7,9, 11,12,13,14	11	2	0
6	2,6,9,12,13	5	1	0
7	0,1,2,3,4,5,6, 7,11,13,14	11	1	0

Lemma 3: The optimal test set (OTS) for detecting all multiple missing-gate faults (MMGFs) in a reversible circuit is also sufficient to detect all single missing-gate faults (SMGFs) of the corresponding circuit.

Proof: Since the *OTS* detects all MMGFs, all faulty blocks B_{ij} under multiple missing-gate fault for any values of i and j ($0 \le i \le j \le N-1$), where B_{ij} is a block containing (j-i+1) consecutive gates of the reversible circuit. This implies that the OTS detects the faulty block B_{ij} for any i, $0 \le i \le N-1$. In other words, the OTS detects all SMGFs as B_{ij} is a block

containing single gate for any i, $0 \le i \le N-1$.

Lemma 4: All detectable repeated gate faults (RGFs) are detected by *OTS*.

Proof: Occurrence of a detectable RGFs implies that a gate under fault is repeated consecutively in the circuit even number of times. Thus, if we denote the repetitions of the faulty gate by a block *B*, then the permutation *P(B)=I* where *I* is the identity permutation. Hence, the proof follows.

4. EXPERIMENTAL RESULTS

We have studied several examples of reversible benchmark circuits [13] as shown in Table 6. Column 1 shows the circuit name, column 2 and column 3 denote the gate count (*N*) and the input size (*n*) respectively. The number of tests for detecting all MMGFs obtained by running the ATPG [22], is shown in column 4. The number of tests for detecting all MMGFs obtained by the proposed algorithm is shown in column 5. Column 6 shows the optimal test set (*OTS*) of the corresponding circuit.

Table 6: Results of the proposed method
(S=Single MGF, M=Multiple MGF, R=Repeated gate fault, S+M+R=all models combined)

Circuit	*N*	*n*	# of tests as in [22]	# of tests as in proposed method S+M+R	*OTS*
			M		
2of5d1	18	6	5	5	{9,4,16,20,35}
4_49tc1	16	4	3	3	{3,15,11}
hwb4tc	17	4	4	4	{1,10,5,9}
rd53d1	12	7	3	3	{124,92,116}
rd53rcmg	30	7	4	4	{110,71,42,125}
mod5addres	21	6	4	4	{28,41,32,19}
5mod5tc	17	6	6	3	{18,5,36}
ham3tc	5	3	2	2	{3,5}

5. CONCLUSIONS

This paper presents an algorithm for deriving an optimal test set (OTS) for testing all multiple missing-gate faults (MMGFs) in a reversible circuit. The same test set is also sufficient to detect all single missing-gate faults (SMGFs) and all detectable repeated gate faults (RGFs) in the circuit.

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